

第四次上机实验题目

1、 中断子程序设计

编写程序，使类型 1CH 的中断向量指向中断处理程序 COUNT，COUNT 统计 1CH 中断次数并存入字变量单元 NUM 中。程序启动后等待用户输入，输入字符 Q 后退出，并将 NUM 值用十六进制形式显示出来。例如 NUM 的内容为 1234h，则在屏幕上显示 1234h。

数据段中至少需要定义以下内容：

- (1) ID db '2186123456' (说明：以学号 2186123456 为例)
- (2) 定义中断次数 NUM 的内存单元

2、 BIOS 和 DOS 中断

编写一个程序，接收从键盘输入的 10 个十进制数字（你的学号），输入回车符则停止输入，然后将这些数字加密后（用 XLAT 指令变换）存入内存缓冲区 BUFFER。加密表为：

输入数字： 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

密码数字： 7, 5, 9, 1, 3, 6, 8, 0, 2, 4

数据段中至少需要定义以下内容：

- (1) ID db '2186123456' (说明：以学号 2186123456 为例)
- (2) BUFFER db 10 dup (?)

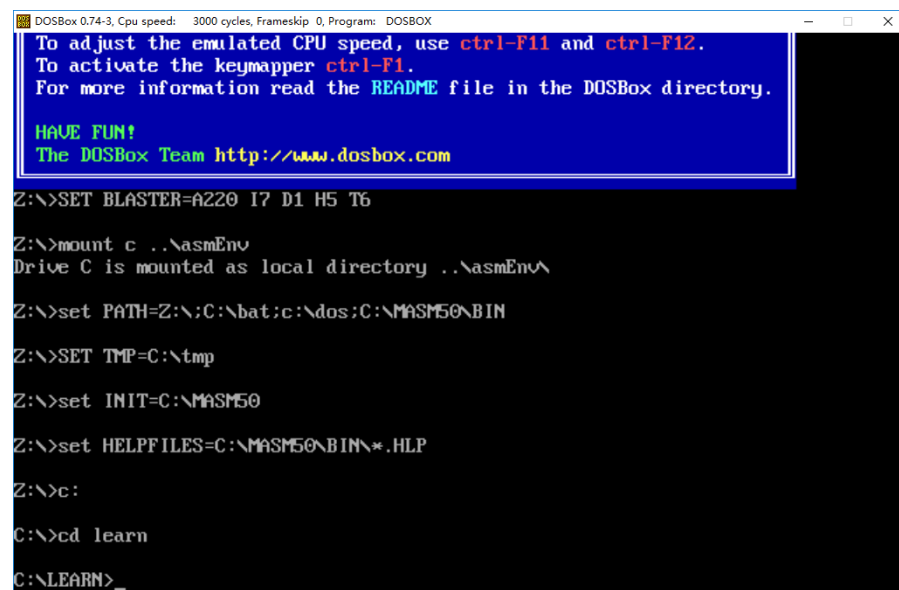
实验要求：

- 1、 在上机时间内，完成上述实验内容，并按要求在思源学堂**提交上机结果**，
具体格式和要求见 [提交上机结果的模板文件](#)；
- 2、 **上机结果文件转换为 PDF 格式后再进行提交。**

截图说明：

在 windows 下使用 **Alt+Print** 组合键可以对当前窗口进行截图，然后可以直接粘贴在这个文档中。这里以上机环境刚启动时为例进行说明，上机环境启动时的

截图：



```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip: 0, Program: DOSBOX
To adjust the emulated CPU speed, use ctrl-F11 and ctrl-F12.
To activate the keymapper ctrl-F1.
For more information read the README file in the DOSBox directory.

HAVE FUN!
The DOSBox Team http://www.dosbox.com

Z:\>SET BLASTER=A220 I7 D1 H5 T6

Z:\>mount c ..\asmEnv
Drive C is mounted as local directory ..\asmEnv\

Z:\>set PATH=Z:\;C:\bat;c:\dos;C:\MASM50\BIN

Z:\>SET TMP=C:\tmp

Z:\>set INIT=C:\MASM50

Z:\>set HELPFILES=C:\MASM50\BIN\*.HLP

Z:\>c:

C:\>cd learn

C:\LEARN>_
```

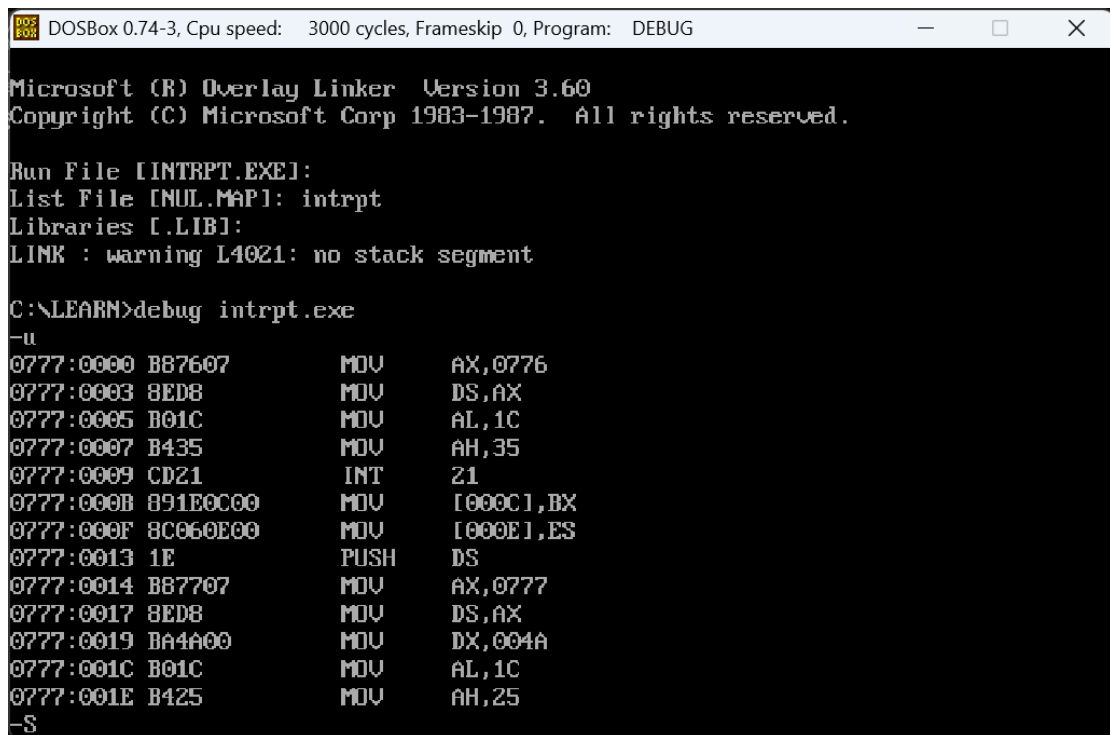
提交上机结果的模板文件

第 4 次上机

班级	学号	姓名
计算机 2301	2234412799	李祥熙

1、中断程序设计

(1) 反汇编的截图



```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
Microsoft (R) Overlay Linker Version 3.60
Copyright (C) Microsoft Corp 1983-1987. All rights reserved.

Run File [INTRPT.EXE]:
List File [NUL.MAP]: intrpt
Libraries [LIB]:
LINK : warning L4021: no stack segment

C:\LEARN>debug intrpt.exe
-u
0777:0000 B87607      MOV     AX,0776
0777:0003 8ED8          MOV     DS,AX
0777:0005 B01C          MOV     AL,1C
0777:0007 B435          MOV     AH,35
0777:0009 CD21          INT     21
0777:000B 891E0C00      MOV     [000C],BX
0777:000F 8C060E00      MOV     [000E],ES
0777:0013 1E          PUSH    DS
0777:0014 B87707      MOV     AX,0777
0777:0017 8ED8          MOV     DS,AX
0777:0019 BA4A00      MOV     DX,004A
0777:001C B01C          MOV     AL,1C
0777:001E B425          MOV     AH,25
-S
```

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
0777:000F 8C060E00 MOV [000E],ES
0777:0013 1E PUSH DS
0777:0014 B87707 MOV AX,0777
0777:0017 8ED8 MOV DS,AX
0777:0019 BA4A00 MOV DX,004A
0777:001C B01C MOV AL,1C
0777:001E B425 MOV AH,25
-u
0777:0020 CD21 INT 21
0777:0022 1F POP DS
0777:0023 B401 MOV AH,01
0777:0025 CD21 INT 21
0777:0027 3C51 CMP AL,51
0777:0029 7406 JZ 0031
0777:002B 3C71 CMP AL,71
0777:002D 7402 JZ 0031
0777:002F EBF2 JMP 0023
0777:0031 1E PUSH DS
0777:0032 8B160C00 MOV DX,[000C]
0777:0036 A10E00 MOV AX,[000E]
0777:0039 8ED8 MOV DS,AX
0777:003B B01C MOV AL,1C
0777:003D B425 MOV AH,25
0777:003F CD21 INT 21
-S
```

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
0777:0031 1E PUSH DS
0777:0032 8B160C00 MOV DX,[000C]
0777:0036 A10E00 MOV AX,[000E]
0777:0039 8ED8 MOV DS,AX
0777:003B B01C MOV AL,1C
0777:003D B425 MOV AH,25
0777:003F CD21 INT 21
-u
0777:0041 1F POP DS
0777:0042 EB1300 CALL 0058
0777:0045 B8004C MOV AX,4C00
0777:0048 CD21 INT 21
0777:004A 50 PUSH AX
0777:004B 1E PUSH DS
0777:004C B87607 MOV AX,0776
0777:004F 8ED8 MOV DS,AX
0777:0051 FF060A00 INC WORD PTR [000A]
0777:0055 1F POP DS
0777:0056 5B POP AX
0777:0057 CF IRET
0777:0058 A10A00 MOV AX,[000A]
0777:005B B90400 MOV CX,0004
0777:005E 51 PUSH CX
0777:005F B104 MOV CL,04
-S_
```

```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
0777:0057 CF      IRET
0777:0058 A10A00    MOV     AX,[000A]
0777:005B B90400    MOV     CX,0004
0777:005E 51      PUSH    CX
0777:005F B104      MOV     CL,04
-u
0777:0061 D3C0      ROL     AX,CL
0777:0063 59      POP     CX
0777:0064 50      PUSH    AX
0777:0065 240F      AND     AL,0F
0777:0067 3C09      CMP     AL,09
0777:0069 7E02      JLE     006D
0777:006B 0407      ADD     AL,07
0777:006D 0430      ADD     AL,30
0777:006F 8AD0      MOV     DL,AL
0777:0071 B402      MOV     AH,02
0777:0073 CD21      INT     21
0777:0075 58      POP     AX
0777:0076 E2E6      LOOP   005E
0777:0078 B268      MOV     DL,68
0777:007A B402      MOV     AH,02
0777:007C CD21      INT     21
0777:007E C3      RET
0777:007F C404      LES     AX,[SI]
-S_

```

(2) 在进行计算前，显示 ID、NUM 的内存值的截图（多显示、少显示均扣分）

```

-d 0 a
0776:0000 32 32 33 34 34 31 32 37-39 39 00      2234412799.

```

(3) 运行到返回 dos 前暂停，对屏幕显示的输出结果（NUM 值的对应的 ASCII 字符串）截图【结果要与步骤（4）中的内存值一致】

```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
0777:003D B425      MOV     AH,25
0777:003F CD21      INT     21
-u
0777:0041 1F      POP     DS
0777:0042 E81300    CALL   0058
0777:0045 B8004C    MOV     AX,4C00
0777:0048 CD21      INT     21
0777:004A 50      PUSH    AX
0777:004B 1E      PUSH    DS
0777:004C B87607    MOV     AX,0776
0777:004F BED8      MOV     DS,AX
0777:0051 FF060A00  INC     WORD PTR [000A]
0777:0055 1F      POP     DS
0777:0056 58      POP     AX
0777:0057 CF      IRET
0777:0058 A10A00    MOV     AX,[000A]
0777:005B B90400    MOV     CX,0004
0777:005E 51      PUSH    CX
0777:005F B104      MOV     CL,04
-g45
Q0020h
AX=0268 BX=1260 CX=0000 DX=1268 SP=0000 BP=0000 SI=0000 DI=0000
DS=0776 ES=F000 SS=0775 CS=0777 IP=0045  NV UP EI PL NZ NA PE NC
0777:0045 B8004C      MOV     AX,4C00
-S

```

(4) 在完成步骤（3）操作后，立即显示 ID、NUM 的内存值的截图（多显示、少

显示均扣分)

```
-d 0 a
0776:0000 32 32 33 34 34 31 32 37-39 39 20                2234412799
```

(5) 源代码

```
title Task 1 - Interrupt
```

```
data segment
```

```
    ID db '2234412799'
```

```
    NUM dw 0
```

```
    OLD_INT1C_OFF dw ?
```

```
    OLD_INT1C_SEG dw ?
```

```
data ends
```

```
code segment
```

```
    assume cs:code, ds:data
```

```
main proc
```

```
    mov ax, seg data
```

```
    mov ds, ax
```

```
    ; Save old 1CH interrupt vector
```

```
    mov al, 1ch
```

```
    mov ah, 35h
```

```
    int 21h
```

```
    mov OLD_INT1C_OFF, bx
```

```
    mov OLD_INT1C_SEG, es
```

```
    ; Set new 1CH interrupt vector
```

```
    push ds
```

```
    mov ax, seg COUNT
```

```
    mov ds, ax
```

```
    mov dx, offset COUNT
```

```
mov al, 1ch
mov ah, 25h
int 21h
pop ds
```

WAIT_INPUT:

```
; Wait for user input
mov ah, 01h
int 21h
cmp al, 'Q'
je EXIT
cmp al, 'q'
je EXIT
jmp WAIT_INPUT
```

EXIT:

```
; Restore old 1CH interrupt vector
push ds
mov dx, OLD_INT1C_OFF
mov ax, OLD_INT1C_SEG
mov ds, ax
mov al, 1ch
mov ah, 25h
int 21h
pop ds

; Display NUM in hex
call PRINT_NUM

; Return to DOS
```

```
    mov ax, 4c00h
    int 21h
main endp
```

```
COUNT proc far
    push ax
    push ds
    mov ax, seg data
    mov ds, ax
    inc NUM
    pop ds
    pop ax
    iret
COUNT endp
```

```
PRINT_NUM proc
    ; Print word in NUM as hex
    mov ax, NUM
    mov cx, 4
PRINT_LOOP:
    push cx
    mov cl, 4
    rol ax, cl
    pop cx
    push ax
    and al, 0fh
    cmp al, 9
    jle PRINT_DIGIT
    add al, 7
PRINT_DIGIT:
```



```
    add al, 30h
    mov dl, al
    mov ah, 02h
    int 21h
    pop ax
    loop PRINT_LOOP

; Print 'h'
    mov dl, 'h'
    mov ah, 02h
    int 21h
    ret
PRINT_NUM endp
code ends
end main
```

3、BIOS 和 DOS 中断

(1) 反汇编的截图

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
0778:0018 D7 XLAT
0778:0019 88850A00 MOV [DI+000A],AL
0778:001D 47 INC DI
0778:001E E2EE LOOP 000E
-debug bios.exe
^ Error
-quit

C:\LEARN>debug bios.exe
-u
0778:0000 B87607 MOV AX,0776
0778:0003 8ED8 MOV DS,AX
0778:0005 B90A00 MOV CX,000A
0778:0008 BF0000 MOV DI,0000
0778:000B BB1400 MOV BX,0014
0778:000E B401 MOV AH,01
0778:0010 CD21 INT 21
0778:0012 3C0D CMP AL,0D
0778:0014 740A JZ 0020
0778:0016 2C30 SUB AL,30
0778:0018 D7 XLAT
0778:0019 88850A00 MOV [DI+000A],AL
0778:001D 47 INC DI
0778:001E E2EE LOOP 000E
-S_
```

```
DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
0778:000B BB1400 MOV BX,0014
0778:000E B401 MOV AH,01
0778:0010 CD21 INT 21
0778:0012 3C0D CMP AL,0D
0778:0014 740A JZ 0020
0778:0016 2C30 SUB AL,30
0778:0018 D7 XLAT
0778:0019 88850A00 MOV [DI+000A],AL
0778:001D 47 INC DI
0778:001E E2EE LOOP 000E
-u
0778:0020 B8004C MOV AX,4C00
0778:0023 CD21 INT 21
0778:0025 00EB ADD BL,CH
0778:0027 04FF ADD AL,FF
0778:0029 867AFF XCHG BH,[BP+SI-01]
0778:002C A17008 MOV AX,[0870]
0778:002F 39867AFF CMP [BP+FF7A],AX
0778:0033 7203 JB 0038
0778:0035 E98D00 JMP 00C5
0778:0038 8A86FAFE MOV AL,[BP+FEFA]
0778:003C 2AE4 SUB AH,AH
0778:003E 40 INC AX
0778:003F 50 PUSH AX
-S_
```

(2) 在进行计算前，显示 ID、BUFFER 的内存值的截图（多显示、少显示均扣分）

```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
0778:0025 00EB      ADD     BL,CH
0778:0027 04FF      ADD     AL,FF
0778:0029 867AFF      XCHG    BH,[BP+SI-01]
0778:002C A17008      MOV     AX,[0870]
0778:002F 39867AFF      CMP     [BP+FF7A],AX
0778:0033 7203      JB      0038
0778:0035 E98D00      JMP     00C5
0778:0038 8A86FAFE      MOV     AL,[BP+FEFA]
0778:003C 2AE4      SUB     AH,AH
0778:003E 40      INC     AX
0778:003F 50      PUSH    AX
-t
AX=0776 BX=0000 CX=0045 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0766 ES=0766 SS=0775 CS=0778 IP=0003  NV UP EI PL NZ NA PO NC
0778:0003 8ED8      MOV     DS,AX
-t
AX=0776 BX=0000 CX=0045 DX=0000 SP=0000 BP=0000 SI=0000 DI=0000
DS=0776 ES=0766 SS=0775 CS=0778 IP=0005  NV UP EI PL NZ NA PO NC
0778:0005 B90A00      MOV     CX,000A
-d 0 13
0776:0000 32 32 33 34 34 31 32 37-39 39 00 00 00 00 00 00  2234412799.....
0776:0010 00 00 00 00                                     ....
-S

```

(3) 输入回车后, 显示 ID、BUFFER 的内存值的截图 (多显示、少显示均扣分)

```

DOSBox 0.74-3, Cpu speed: 3000 cycles, Frameskip 0, Program: DEBUG
0778:001D 47      INC     DI
0778:001E E2EE      LOOP    000E
-u
0778:0020 B8004C      MOV     AX,4C00
0778:0023 CD21      INT     21
0778:0025 00EB      ADD     BL,CH
0778:0027 04FF      ADD     AL,FF
0778:0029 867AFF      XCHG    BH,[BP+SI-01]
0778:002C A17008      MOV     AX,[0870]
0778:002F 39867AFF      CMP     [BP+FF7A],AX
0778:0033 7203      JB      0038
0778:0035 E98D00      JMP     00C5
0778:0038 8A86FAFE      MOV     AL,[BP+FEFA]
0778:003C 2AE4      SUB     AH,AH
0778:003E 40      INC     AX
0778:003F 50      PUSH    AX
-g20
2234412799
AX=0104 BX=0014 CX=0000 DX=0000 SP=0000 BP=0000 SI=0000 DI=000A
DS=0776 ES=0766 SS=0775 CS=0778 IP=0020  NV UP EI PL NZ NA PE NC
0778:0020 B8004C      MOV     AX,4C00
-d 0 13
0776:0000 32 32 33 34 34 31 32 37-39 39 09 09 01 03 03 05  2234412799.....
0776:0010 09 00 04 04                                     ....
-S_

```

(4) 源代码

title Task 2 - BIOS and DOS Interrupt

data segment

ID db '2234412799'

```
    BUFFER db 10 dup (?)
    TABLE db 7, 5, 9, 1, 3, 6, 8, 0, 2, 4
data ends
```

code segment

```
    assume cs:code, ds:data
```

```
main proc
```

```
    mov ax, data
```

```
    mov ds, ax
```

```
    mov cx, 10
```

```
    mov di, 0
```

```
    mov bx, offset TABLE
```

```
READ_LOOP:
```

```
    ; Read char from keyboard
```

```
    mov ah, 01h
```

```
    int 21h
```

```
    ; Check for Enter (0Dh)
```

```
    cmp al, 0dh
```

```
    je END_INPUT
```

```
    ; Convert ASCII to number
```

```
    sub al, 30h
```

```
    ; Encrypt using XLAT
```

```
    xlat
```

```
    ; Store to BUFFER
```

```
    mov BUFFER[di], al
    inc di
    loop READ_LOOP
```

```
END_INPUT:
```

```
    ; Return to DOS
```

```
    mov ax, 4c00h
```

```
    int 21h
```

```
main endp
```

```
code ends
```

```
end main
```